Four-perfboard hybrid — assessment

Two contact banks above; power and controller/matrix below on a removable tray.

Not released for soldering. Software checks do not establish measured fit, manual-wire safety, protection response or PCB manufacturability.

Gate	Result	Meaning
Software / net preservation	pass	1659 original connected pins compared
Measured mechanical fit	pending	Actual board grid, adapter bodies, stack height and extraction pending
Powered qualification	pending	No measured hardware acceptance supplied
Smart PCB fabrication	pending	Reference contracts only; no routed fabrication files
Overall decision	pending	Four-board target retained; no extra lower board permitted

Assembly	Role	Local wires	Functional parts
contact-a	Bank A — continuous contact network	97	20
contact-b	Bank B — continuous contact network	97	20
power	Power — input, guards, student cutoff and interlock	167	64
router-matrix	Router — ESP1, watchdog, matrices and Pi interfaces	219	63

Smart circuitry remains separate: the 12 inherited layouts are **logical electrical contracts**, not twelve proposed PCBs or extra perfboards.

Construction effort — compare the same scope

Metric	R5 retained groups	R6 permanent boards
Perfboards	15 (contacts + power + router + interfaces)	4
Local wire segments	597	580
Functional parts	167	167
SMD adapters	9	9
Board solder estimate	See R5 whole-system report	2023

The whole R5 system has 27 perfboards and 1,392 local wires. Moving the smart subsystem to PCB removes its 795 local hand-wire segments from the permanent perfboard scope. Consolidating the remaining groups removes only 17 local segments: **board count falls much faster than wiring effort**.Current boundary harness model: 210 conductors / 420 ends, versus 279 / 558 in R5. These include inherited seams inside the smart reference contracts. Final PCB consolidation changes the harness count; this is not a final cable purchase list. Solder estimates exclude socket/adapter preparation, carrier internals and harness ends.

Board	Wires	Segment crossings	Wire length	Longest sensitive path
contact-a	97	49	3.09 m	0 mm
contact-b	97	23	3.08 m	0 mm
power	167	854	8.56 m	86.4 mm
router-matrix	219	2259	14.17 m	66 mm

Crossings count strict segment intersections between different nets. They are a congestion indicator, not shorts or a complete overlap count. Insulation, lead access and analog stability require physical inspection.

Mechanical fit and removable tray

Contact banks

- 11 columns per bank; 177.80 mm first-to-last span.
- Six contacts per column; 50.80 mm row span.
- Assumed 78 × 38 grid at 2.54 mm; check actual stock.
- Capsule orientation is unchanged between banks.
- Contact frame carries insertion force independently.

Lower boards

- Power zones: input/guards and student cutoff/interlock.
- Router zones: ESP1/watchdog and matrices/control/interfaces.
- Module, cable and smart PCB space is additional to pad packing.

Required clearance — fill from measurements Contact underside protrusions + tallest lower-board assembly + lower-board thickness + underside joints + insulation allowance + cable bend envelope.
Calculated requirement: **pending**. Available height: **pending**.
Use measurements.json, component-envelope-review.csv and manual-review.json. The required calculation is conservative; verify routing-specific clearances in the mock-up.

Print the scale templates at 100%, measure the calibration bar, place actual sockets/modules over the outlines, and rehearse tray removal. If two lower boards cannot provide fit and repair access, mark the arrangement failed; report the matrix functions that need PCB consolidation.

Smart prototype — staged evaluation

Not released for soldering. Software checks do not establish measured fit, manual-wire safety, protection response or PCB manufacturability.

Candidate	Result	Consequence
Complete R5 one-position support on 7×10	full-7x10-layout-failed	smart-reference: U_SMART_SAFETY_USB_POR does not fit 26x38
ESP32 + direct precision voltage front end	provisional-pad-fit	Provisional pad fit only; no mux or isolated multi-position service bus

The early board supports display/voltage work. Continuity stimulus uses the existing bounded eight-part diagnostic circuit as an external removable fixture. Later output/current qualification requires an external temporary fixture or final smart PCB. Do not purchase or solder a full 7×10 prototype from the failed packing.

Standalone signal	Connection / condition
OLED / ADS1115 I2C	GPIO4 SDA; GPIO22 SCL; 0x3C OLED and 0x48 ADC
External continuity stimulus	GPIO26 through removable control lead; OFF at reset; remove before register-latch use
Precision ADC	AIN0 → P0_1_MUX; AIN1 → P0_2_MUX; no CD4052 on this one-channel fixture
Temporary supply	Verified ESP USB3V3 → 3V3_SYS removable lead, never paralleled with final regulator
Final handoff	GPIO26 returns to ESP2 latch; final mux, bus isolation and watchdog tests repeated

Prototype evidence does not pass the permanent assembly checkpoints. Keep its serial number and firmware profile separate. The ADC carrier and OLED are external modules whose mounting envelopes are unverified.

Checkpoint chapters and release decision

Chapter	Checkpoints	Filmable acceptance
First light and diagnostics	CP00–03	LED on/OFF; open and wrong jumpers rejected; correction needs a fresh test
One smart display and voltage	CP04–08	Identity/OFF status; calibration; actual and signed voltage
Final power protection	CP09–14	Qualified rails; startup OFF; latched overload; heartbeat and transient cutoff
One complete smart position	CP15–23	Bounded source/sink, pulse, gates, memory and measured current
Programmable matrix	CP24–26	Routed LED loop, continuity and invalid-route rejection
All positions	CP27–31	Every position qualified; at most three active; removal/reconfiguration safe
System sessions	CP32–36	Pi, explicit enable, RFID failures and complete lesson

CP00 needs no final power or matrix. CP01 diagnostics use fixed connections. CP09 begins power; CP24 begins matrix. Preserve permanent work; only labelled temporary links and test arrangements change. Use checkpoint-guide.pdf and transitions.csv.**October milestones:** October 22 demonstration must remain independent of PCB arrival. October 29 requires tested electronics with partial firmware integration. PCB procurement and firmware work proceed in parallel; missing hardware, incomplete PCB files or unavailable diagnostic firmware remain schedule blockers.**Release rule:** fail on net/placement errors or measured fit failure. Otherwise remain pending until physical access, all applicable bench tests and final PCB manufacturing review have evidence. No computed drawing becomes a measured pass.